perspective they are living life in hyper-slow motion and therefore one of their normal life cycles takes an enormous amount of our time.

Who Is Moving, Anyway?

The relativity of motion is both the key to understanding Einstein's theory and a potential source of confusion. You may have noticed that a reversal of perspective interchanges the roles of the "moving" muons, whose watches we have argued run slowly, and their "stationary" counterparts. Just as both George and Gracie had an equal right to declare that they were stationary and that the other was moving, the muons we have described as being in motion are fully justified in proclaiming that, from their perspective, they are motionless and that it is the "stationary" muons that are moving, in the opposite direction. The arguments presented can be applied equally well from this perspective, leading to the seemingly opposite conclusion that watches worn by the muons we christened as stationary are running slow compared with those worn by the muons we described as moving.

We have already met a situation, the signing ceremony with the light bulb, in which different viewpoints lead to results that seem to be completely at odds. In that case we were forced by the basic reasoning of special relativity to give up the ingrained idea that everyone, regardless of state of motion, agrees about which events happen at the same time. The present incongruity, though, appears to be worse. How can two observers each claim that the other's watch is running slower? More dramatically, the different but equally valid muon perspectives seem to lead us to the conclusion that each group will claim that it is the other group that dies first. We are learning that the world can have some unexpectedly strange features, but we would hope that it does not cross into the realm of logical absurdity. So what's going on?

As with all apparent paradoxes arising from special relativity, under close examination these logical dilemmas resolve to reveal new insights into the workings of the universe. To avoid ever more severe anthropomorphizing, let's switch from muons back to George and Gracie, who now, in addition to their flashing lights, have bright digital clocks on their spacesuits. From George's perspective, he is stationary while Gracie with her flashing green light and large digital clock appears in the distance and then passes him in the blackness of empty